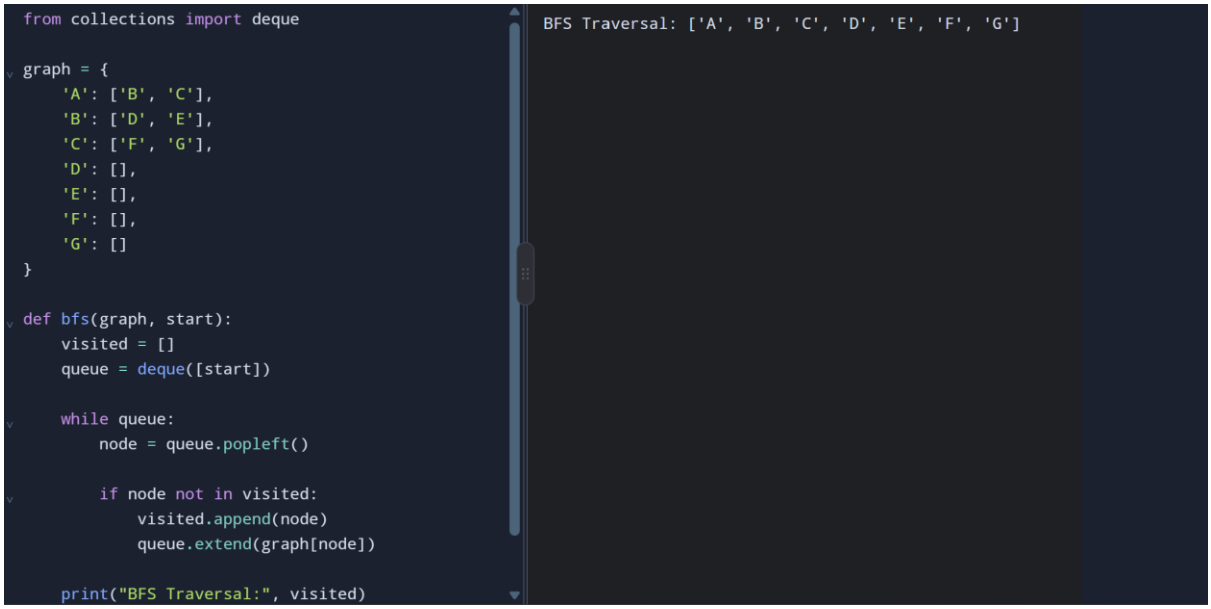
BFS Programs

1



```python
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [],
    'E': [],
    'F': [],
    'G': []
}

def bfs(graph, start):
    visited = []
    queue = deque([start])

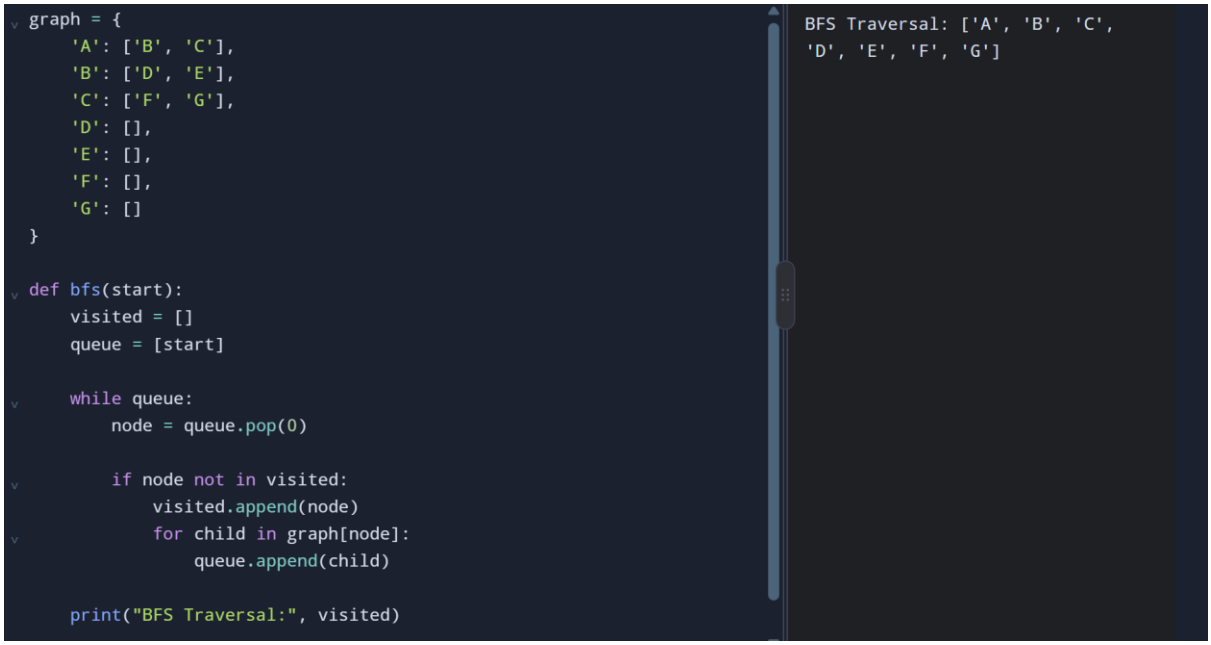
    while queue:
        node = queue.popleft()

        if node not in visited:
            visited.append(node)
            queue.extend(graph[node])

    print("BFS Traversal:", visited)
```

```
BFS Traversal: ['A', 'B', 'C', 'D', 'E', 'F', 'G']
```

2



```python
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [],
    'E': [],
    'F': [],
    'G': []
}

def bfs(start):
    visited = []
    queue = [start]

    while queue:
        node = queue.pop(0)

        if node not in visited:
            visited.append(node)
            for child in graph[node]:
                queue.append(child)

    print("BFS Traversal:", visited)
```

```
BFS Traversal: ['A', 'B', 'C',
'D', 'E', 'F', 'G']
```

3

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [], 'E': [], 'F': [], 'G': []
}

level = ['A']

while level:
    print(*level, end=" ")
    next_level = []

    for node in level:
        next_level.extend(graph[node])

    level = next_level
```

```
A B C D E F G
```

4

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [], 'E': [], 'F': [], 'G': []
}

def bfs(start):
    visited = {start}
    queue = deque([start])

    while queue:
        node = queue.popleft()
        print(node, end=" ")

        for child in graph[node]:
            if child not in visited:
                visited.add(child)
                queue.append(child)

```

```
A B C D E F G
```